ECONOMETRICS I
Practice Problem Set 4
(chapters 7-8-9)

1. In order to find the determinants of time spent sleeping, the following regression model is estimated by using a random sample of 706 workers:

$$\widehat{\text{sleep}} = \underset{(235.1)}{3840} - \underset{(0.018)}{0.163}\,\text{totwrk} - \underset{(5.85)}{11.71}\,\text{educ} - \underset{(11.20)}{8.70}\,\text{age} + \underset{(0.12)}{0.13}\,age^2 + \underset{(34.32)}{87.75}\,\text{male}$$

$$n = 706 \quad R^2 = 0.122$$

(standard errors in parentheses)

Variable definitions: ***sleep*:** total <u>minutes</u> per week spent sleeping, ***totwrk*:** total weekly minutes spent working, ***educ*:** education level measured in years, ***age*:** a person's age measured in years, age^2**:** age-squared, **male:** dummy variable taking the value 1 if the worker is male and 0 otherwise.

   (a) The base group in the model:.....
   (b) The intercept for male is estimated as .....; whereas for female it is .....
   (c) Interpret the estimated coefficient on *male*.
   (d) If someone works 6 more <u>hours</u> per week, by how many minutes is *sleep* predicted to fall, ceteris paribus?
   (e) Test the null hypothesis $H_0 : \beta_{educ} = -12$ at 5% level.
   (f) If *age* increases 25 to 30, find the estimated change in *sleep*.
   (g) Describe the relationship between *sleep* and *age*? Find the age at which this relationship becomes positive.
   (h) Test the overall significance of the above regression at 5%.
   (i) Suppose that an important variable is excluded in this model. Give an example of such situation in the context of the model and explain which assumption might be violated and why?
   (j) Suppose that the normality assumption is violated in the sleep regression, briefly discuss whether this violation is a critical issue or not.
   (k) Suppose that the sleep regression contains heteroscedasticity. How does this affect the test results in this question?

2. The following equations were estimated using the data in BWGHT:

$$\widehat{\log(bwght)} = \underset{(.22)}{4.66} - \underset{(.0009)}{.0044}\,cigs + \underset{(.0059)}{.0093}\log(faminc) + \underset{(.006)}{.016}\,parity + \underset{(.010)}{.027}\,male + \underset{(.013)}{.055}\,white$$

$$n = 1388, R^2 = .0472.$$

and

$$\widehat{\log(bwght)} = \underset{(.38)}{4.65} - \underset{(.0010)}{.0052}\,cigs + \underset{(.0085)}{.0110}\log(faminc) + \underset{(.006)}{.017}\,parity + \underset{(.011)}{.034}\,male + \underset{(.015)}{.045}\,white - \underset{(.0030)}{.0030}\,motheduc + \underset{(.0026)}{.0032}\,fatheduc$$

$$n = 1191, R^2 = .0493.$$

The variables are defined as in Example 4.9, but we have added a dummy variable for whether the child is male and a dummy variable indicating whether the child is classified as white.

(a) In the first equation, interpret the coefficient on the variable *cigs*. In particular, what is the effect on birth weight from smoking 10 more cigarettes per day?

(b) How much more is a white child predicted to weigh than a nonwhite child, holding the other factors in the first equation fixed? Is the difference statistically significant?

(c) Comment on the estimated effect and statistical significance of *motheduc*.

(d) From the given information, why are you unable to compute the F statistic for joint significance of *motheduc* and *fatheduc*? What would you have to do to compute the F statistic?

3. Using the data in GPA2, the following equation was estimated:

$$\begin{aligned}\widehat{sat} &= \underset{(6.29)}{1028.10} + \underset{(3.83)}{19.30}hsize - \underset{(.53)}{2.19}hsize^2 - \underset{(4.29)}{45.09}female \\ &\quad - \underset{(12.71)}{169.81}black + \underset{(18.15)}{62.31}female \cdot black \\ n &= 4137, R^2 = .0858.\end{aligned}$$

The variable *sat* is the combined SAT score; *hsize* is size of the student?s high school graduating class, in hundreds; *female* is a gender dummy variable; and *black* is a race dummy variable equal to one for blacks, and zero otherwise.

(a) Is there strong evidence that $hsize^2$ should be included in the model? From this equation, what is the optimal high school size?

(b) Holding *hsize* fixed, what is the estimated difference in SAT score between nonblack females and nonblack males? How statistically significant is this estimated difference?

(c) What is the estimated difference in SAT score between nonblack males and black males? Test the null hypothesis that there is no difference between their scores, against the alternative that there is a difference.

(d) What is the estimated difference in SAT score between black females and nonblack females? What would you need to do to test whether the difference is statistically significant?

4. An equation explaining chief executive officer salary is

$$\begin{aligned}\widehat{\log(salary)} &= \underset{(.30)}{4.59} + \underset{(0.32)}{.257}\log(sales) + \underset{(.004)}{.011}roe + \underset{(.89)}{.158}finance \\ &\quad + \underset{(.085)}{.181}consprod - \underset{(.099)}{.283}utility \\ n &= 209, R^2 = .357.\end{aligned}$$

The data used are in CEOSAL1, where *finance*, *consprod*, and *utility* are binary variables indicating the financial, consumer products, and utilities industries. The omitted industry is transportation.

(a) Compute the approximate percentage difference in estimated salary between the utility and transportation industries, holding *sales* and *roe* fixed. Is the difference statistically significant at the 1% level?

(b) Use equation (7.10) to obtain the exact percentage difference in estimated salary between the utility and transportation industries and compare this with the answer obtained in part (a).

(c) What is the approximate percentage difference in estimated salary between the consumer products and finance industries? Write an equation that would allow you to test whether the difference is statistically significant.

5. Consider the following model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

where x_1 is related to the error variance as follows:

$$Var(u|x_1, x_2) = \sigma^2 x_1$$

where $x_1 > 0$. Assume that the rest of the Gauss-Markov assumptions hold.

   (a) What are the properties of OLS estimators? Are they still consistent? Are they the most efficient?

   (b) Find the GLS transformation and describe the steps of GLS estimation.

6. Consider the following regression results for sleep based on a random sample of 706 workers:

$$\widehat{\text{sleep}} = \underset{(235.1)}{3840} - \underset{(0.018)}{0.163}\,\text{totwrk} - \underset{(5.85)}{11.71}\,\text{educ} - \underset{(11.20)}{8.70}\,\text{age} + \underset{(0.12)}{0.13}\,age^2 + \underset{(34.32)}{87.75}\,\text{male}$$

$$n = 706 \quad R^2 = 0.122$$

(standard errors in parentheses)

Variable definitions: ***sleep*:** total minutes per week spent sleeping, ***totwrk*:** total weekly minutes spent working, ***educ*:** education level measured in years, ***age*:** a person's age measured in years, age^2**:** age-squared, **male:** dummy variable taking the value 1 if the worker is male and 0 otherwise.

   (a) Suppose that the sleep regression contains heteroscedasticity. How does this affect the t and F test results?

   (b) Squared Residuals are obtained from the regression above and then regressed on all explanatory variables The coefficient of determination from this auxiliary regression was found to be $R^2_{\hat{u}^2} = 0.02$. Test whether there is heteroscedasticity in this model. State which test you used, the null hypothesis and your decision. (critical value is 11.07 at 5% significance level).

   (c) The researcher suspects that there may be functional form misspecification in this model and computes the RESET test statistic. The p-value associated with the RESET test is found to be 0.042. Does the functional form seem to be correct? State the null hypothesis and your decision at 5% significance level.

   (d) If we re-estimate the model for only men (male=1 group) would OLS estimators be still consistent? Is this endogenous or exogenous sample selection?

7. Wooldridge text ex. 8.2 on p.268

8. Wooldridge text ex. 8.5 on p.269

9. Wooldridge text ex. 9.1 on p.303

10. Wooldridge text ex. 9.4 on p.304